

1.

- a. Dense compute
 - i. N^2 MACs in a $N \times N$ MAC matrix, 2 FLOPs per MAC
 - ii. Dense FLOPs = $2N^2 = 2 * 512^2 = 524,288$ flops
- b. Dense bytes = $4N^2 = 4 * 512^2 = 1,048,576$
- c. Sparse compute
 - i. S is fraction of weights that are 0, $1-s$ is non-zero weights
 - ii. Sparse flops = $2N^2 * (1-s)$
- d. Sparse bytes
 - i. Store every nonzero weight, each size 4 bytes, and its column location
 - ii. $= 4(1-s)N^2 * 2$
 - iii. Store starting position of every row and final row pointer, each 4 bytes
 - iv. $= 4(N+1)$
 - v. Total sparse bytes = $8(1-s)N^2 + 4(N+1) = 2,097,152(1-s) + 2052$

2. Theoretical speedup of sparse vs dense MVM

- a. Speedup = Dense FLOPs / Sparse FLOPs = $2N^2 / (2(1-s)N^2) = 1 / (1-s)$
- b. Speedup = 2 = $1 / (1-s)$
- c. $S = 0.5$

3. Memory tradeoff

- a. Set sparse memory equal to dense memory
- b. $8(1-s)N^2 + 4(N+1) = 4N^2$
- c. $8(1-s)N^2 = 4N^2 - 4(n+1)$
- d. $1-s = (4N^2 - 4(n+1)) / 8N^2 = (N^2 - N - 1) / 2N^2$
- e. $S = 1 - ((N^2 - N - 1) / 2N^2)$
- f. $S = 1 - ((512^2 - 512 - 1) / 2 * 512^2) = 1 - 261631 / 524288 = 1 - .49902 = 0.50098$
- g. $S = 50.0098\%$

4. Find end-to-end speedup

- a. Dense bytes = $4N^2 = 1,048,576$
- b. Sparse bytes = $8(1-s)N^2 + 4(n+1) = 8(0.1)(262,144) + 2052 = 211,767.2$ bytes
- c. Memory bandwidth = 320 GB/s = $320 * 10^9$ bytes/s
- d. Dense time = bytes / memory = $1,048,576 / 320 * 10^9 = 3.28$ us
- e. Sparse time = bytes / memory = $211,762.2 / 320 * 10^9 = 0.662$ us
- f. Speedup = dense time / sparse time = $3.28 / 0.662 = 4.95$
- g. 4.95x faster